# Rheumatology in Critical Care – Chapter Summary

Vasculitis • CTD • Pulmonary-Renal • Catastrophic Syndromes

**ICU / EDIC Board Review • MCQ-Oriented**

## 1. Approach in the ICU

- Presentations: **pulmonary-renal syndrome**, DAH, myocarditis, mesenteric ischemia, CNS vasculitis, TMA
- **Always exclude infection** first (mimic & complication of immunosuppression)
- Drug toxicity (methotrexate pneumonitis, cyclophosphamide cystitis)
- Early **plasma exchange / pulse steroids** may be life-saving

## 2. Autoantibodies

| Antibody | Disease |
|---|---|
| c-ANCA (PR3) | Granulomatosis with polyangiitis (GPA) |
| p-ANCA (MPO) | Microscopic polyangiitis, EGPA |
| Anti-GBM | Goodpasture (anti-GBM disease) |
| Anti-dsDNA, anti-Sm | SLE (dsDNA tracks activity, ↓ C3/C4) |
| Anti-Scl-70 | Diffuse systemic sclerosis |
| Anti-centromere | Limited SSc (CREST) |
| Anti-Jo-1 | Antisynthetase syndrome (ILD, myositis) |
| Anti-MDA5 | Amyopathic dermatomyositis – rapidly progressive ILD |
| Antiphospholipid | APS (lupus anticoagulant, anticardiolipin, β2GP1) |
| Anti-CCP | Rheumatoid arthritis |

## 3. Pulmonary-Renal Syndrome

- **DAH + rapidly progressive GN**
- **Causes:** ANCA vasculitis (most common), anti-GBM, SLE, APS, TTP
- **Workup:** urinalysis (RBC casts), ANCA, anti-GBM, ANA, complement, BAL (progressively bloodier), renal biopsy
- **Rx:** pulse methylprednisolone 0.5–1 g × 3 days + **rituximab** or cyclophosphamide
- **Plasma exchange:** anti-GBM (definitive); ANCA – PEXIVAS: no mortality / ESRD benefit (selected severe cases)
- **Avacopan** (C5aR inhibitor) allows steroid reduction (ADVOCATE)

## 4. ANCA-Associated Vasculitis

- **GPA:** upper airway (saddle nose, sinusitis), cavitating nodules, GN, subglottic stenosis
- **MPA:** GN + alveolar hemorrhage; no granulomas
- **EGPA:** asthma + eosinophilia + neuropathy; cardiac involvement = poor prognosis → mepolizumab
- **Five Factor Score** for prognosis

## 5. Systemic Lupus Erythematosus

- ICU causes: infection (most common), lupus flare, nephritis, serositis, DAH, CNS lupus, myocarditis, TMA
- **Flare vs infection:** ↓ complement & ↑ dsDNA favor flare; **CRP often normal in flare** (high CRP → infection)
- **Rx:** methylprednisolone pulse, cyclophosphamide / mycophenolate, rituximab, belimumab; hydroxychloroquine continued
- **Lupus nephritis** class III/IV → induction therapy
- **Macrophage activation syndrome** (MAS / HLH)

## 6. Catastrophic APS (CAPS)

- ≥ 3 organs involved within 1 week, small-vessel thrombosis, antiphospholipid antibodies
- Triggers: infection, surgery, anticoagulation withdrawal
- **Triple therapy:** anticoagulation (heparin) + glucocorticoids + plasma exchange and/or IVIG
- Refractory: rituximab, eculizumab
- Mortality ~ 30–50%
- DOACs inferior to warfarin in high-risk APS

## 7. Systemic Sclerosis

- **Scleroderma renal crisis:** malignant HTN + AKI, TMA; risk ↑ with steroids (> 15 mg/day)
- **Rx: ACE inhibitor (captopril)** – even if creatinine rising
- **PAH / ILD** major causes of death
- Difficult airway (microstomia), GI dysmotility, aspiration

## 8. Inflammatory Myopathies

- **Dermatomyositis / polymyositis:** proximal weakness, ↑ CK, respiratory muscle failure, aspiration, ILD
- **Anti-MDA5:** rapidly progressive ILD, high mortality → triple therapy (steroids + CNI + cyclophosphamide)
- **Antisynthetase:** ILD + myositis + mechanic's hands + arthritis
- Screen for malignancy

## 9. Rheumatoid Arthritis

- **Atlantoaxial subluxation:** careful neck manipulation during intubation
- **Cricoarytenoid arthritis:** stridor, difficult airway
- ILD (UIP pattern), pleural effusion (low glucose)
- Drug toxicity: methotrexate pneumonitis / marrow, anti-TNF infections (TB reactivation)
- Felty syndrome: RA + splenomegaly + neutropenia

## 10. Large-Vessel & Other Vasculitis

- **Giant cell arteritis:** > 50 yrs, headache, jaw claudication, visual loss → immediate steroids (don't wait for biopsy); tocilizumab
- **Takayasu:** young women, absent pulses, aortic involvement
- **Polyarteritis nodosa:** HBV-associated, mesenteric / renal aneurysms, ANCA negative, spares lungs
- **Behçet:** oral / genital ulcers, uveitis, pulmonary artery aneurysms (hemoptysis), venous thrombosis
- **Kawasaki:** coronary aneurysms in children → IVIG + aspirin
- **IgA vasculitis:** purpura, arthritis, GI, nephritis

## 11. Adult-Onset Still's & HLH

- **Still's:** quotidian fever, salmon rash, arthritis, ferritin ↑↑ (glycosylated < 20%)
- **HLH / MAS:** fever, cytopenias, splenomegaly, ferritin > 10,000, ↓ fibrinogen, ↑ TG, ↑ sIL-2R
- **HScore** > 169 suggests HLH
- **Rx:** dexamethasone / methylprednisolone, **anakinra**, IVIG, etoposide (HLH-94); treat trigger

## 12. Drugs & Monitoring

| Drug | Key ICU issues |
|---|---|
| Cyclophosphamide | Hemorrhagic cystitis (mesna), marrow, infertility, PCP prophylaxis |
| Rituximab | Hypogammaglobulinemia, HBV reactivation, PML |
| Methotrexate | Pneumonitis, myelosuppression, ↓ renal clearance |
| Anti-TNF | TB, fungal infection |
| Tocilizumab | CRP suppressed – masks sepsis |
| Hydroxychloroquine | QT prolongation, cardiomyopathy |
| Steroids | Adrenal suppression → stress dose |

- **PCP prophylaxis** with high-dose steroids + cytotoxics

## 13. Crystal & Septic Arthritis

- **Septic arthritis:** S. aureus; aspirate → WBC > 50,000; washout + antibiotics
- **Gout flare** in ICU: colchicine (renal dose), steroids; avoid NSAIDs in AKI
- **Pseudogout:** CPPD, rhomboid positively birefringent crystals

## 14. High-Yield MCQ Pearls

- ☑ Exclude infection before escalating immunosuppression
- ☑ c-ANCA = GPA; p-ANCA = MPA / EGPA
- ☑ Anti-GBM → plasma exchange
- ☑ PEXIVAS: PLEX no benefit routinely in ANCA vasculitis
- ☑ SLE flare: ↓ C3/C4, ↑ dsDNA, CRP normal
- ☑ CAPS: anticoagulation + steroids + PLEX/IVIG
- ☑ Scleroderma renal crisis → ACE inhibitor
- ☑ Anti-MDA5 → rapidly progressive ILD
- ☑ RA: atlantoaxial instability at intubation
- ☑ GCA: steroids before biopsy
- ☑ HLH: ferritin > 10,000, HScore
- ☑ Tocilizumab suppresses CRP

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